Rotman

The session will start at 5:05PM

RITCxSmith 2026 Training Session

November 13, 2026

Rotman School of Management
UNIVERSITY OF TORONTO

Agenda

Rotman

Section	Time
Resource Page	**10 min**
RIT	**45 min**
Platform	25 min
API Connection	20 min
Cases	**60 min**
Volatility Trading	30 min
Algorithmic ETF Arbitrage	30 min
Q&A	**5 min**

RESOURCE PAGE

Practice session and server

Windows / Mac setup

How to connect

How scoring works

Software Setup

Rotman

Windows

Desktop application

Download the RIT User Application

Run the installer, then open RIT.

Server: flserver.rotman.utoronto.ca

Port: 16510



Mac

Browser or Mac app

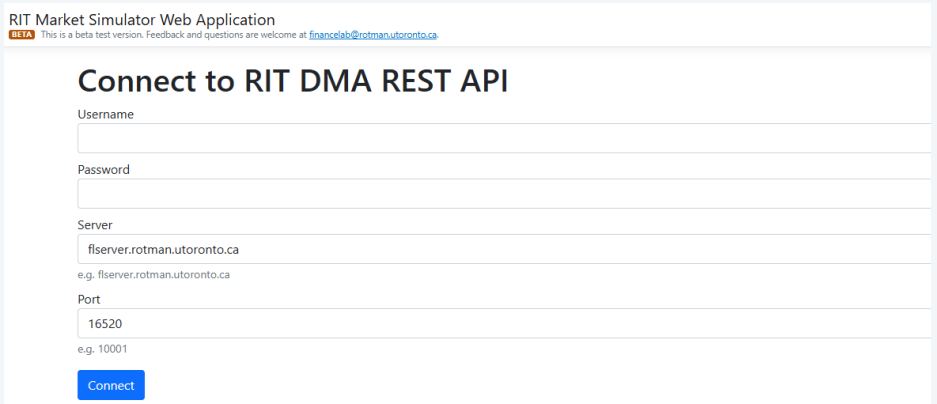
Browser: client.rotmanrit.com:14980

Or download the Apple Silicon / Intel app.

Unzip → drag to Applications → open.

Server: flserver.rotman.utoronto.ca

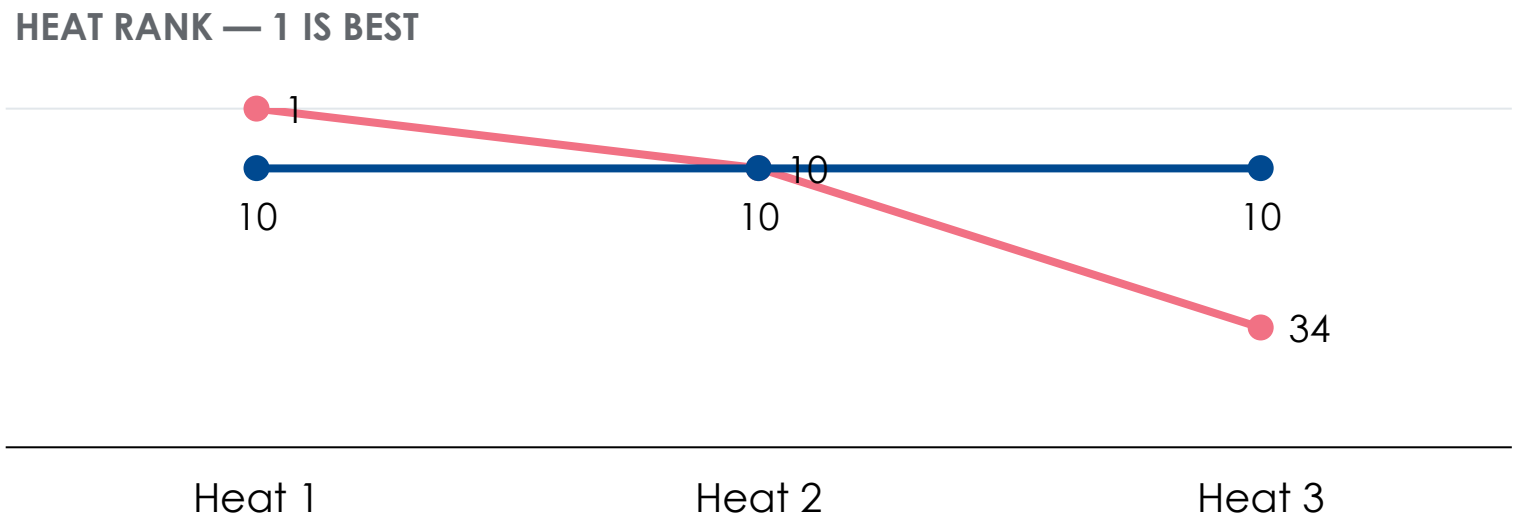
Port: 16520



Scoring

Rotman

41 teams: compare these two teams across three example heats.



AVERAGE HEAT RANK

Team A: 15

Team B: 10

Team B ranks higher.

Lower average is better.

Rank each heat → Average the ranks → Rank each case

Final score: 50% Volatility case score + 50% ETF case score (higher is better)

RIT User Application

Rotman

Read the market, place an instruction, check the result

PORTFOLIO

What do I own?

Position, cost, last, bid and ask.

MARKET DEPTH

What can trade now?

Price, quantity and priority.

ORDER ENTRY / BLOTTER

What did I instruct?

Market / limit; live, partial, filled, cancelled.

TRADER INFO / LOG

What changed?

Limits, P&L, transaction costs; save workspace.

API connection

Rotman

DMA REST API

Python → DMA-enabled simulation server

No desktop client bridge

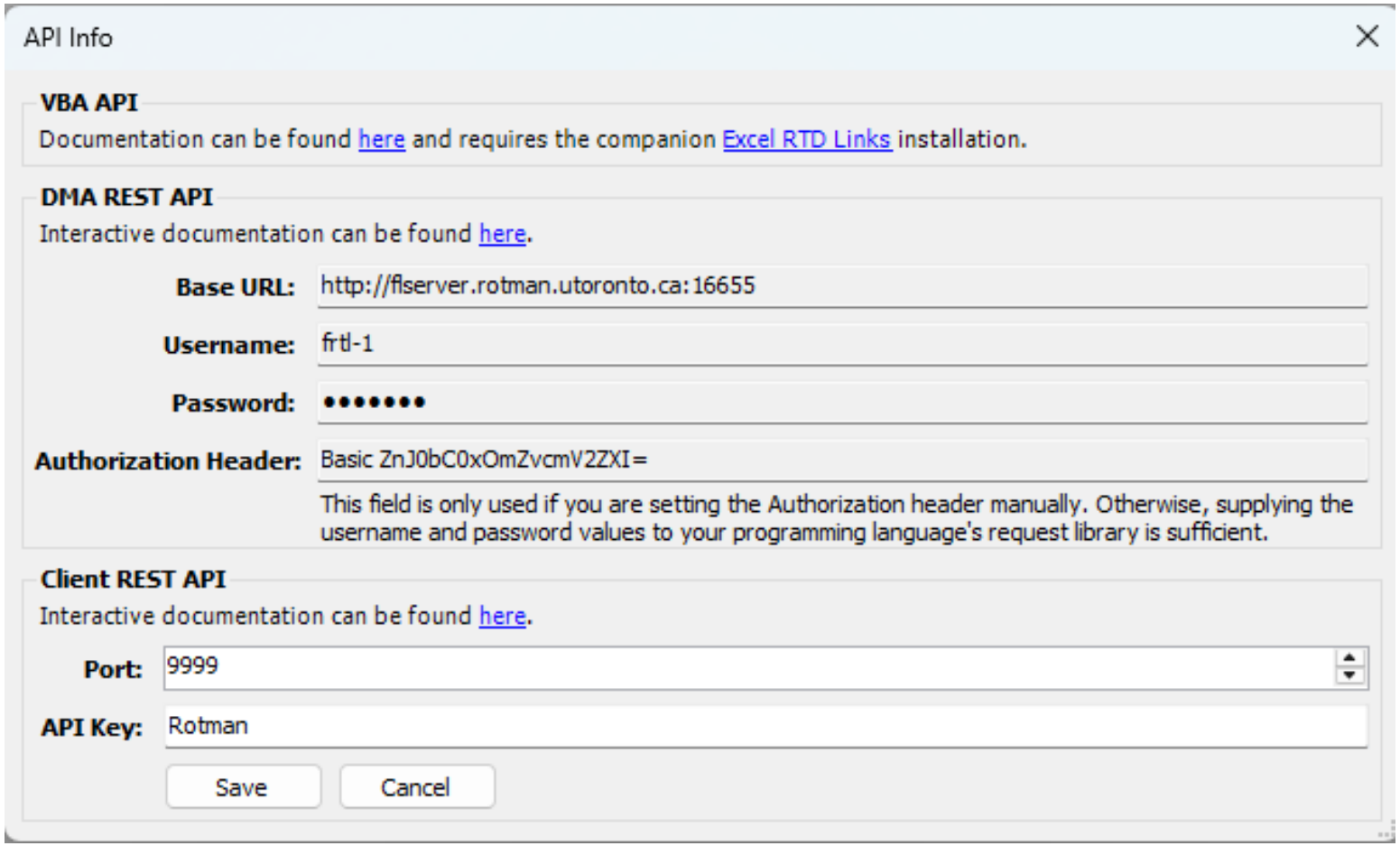
DMA endpoint + trader login

CLIENT REST API

Python → local RIT client → simulation server

Client open and logged in

Local port + matching API key



Volatility Trading Case

Rotman

Black-Scholes: option price and implied volatility

A model price for a European option, given its inputs and assumptions

Stock price (S) Strike (K) Time (T) Interest rate (r)

PRICING AN OPTION

Choose volatility (σ)

Black–Scholes calculates the option price

FINDING IMPLIED VOLATILITY

Start with the market option price

Solve for σ that matches that price

European call option on a non-dividend-paying stock

$$C = S\,N(d_1) - K\,e^{-rT}\,N(d_2)$$

$$d_1 = [\ln(S/K) + (r + \sigma^2/2)T] \,/\, (\sigma\sqrt{T})$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

Volatility Trading Case

Rotman

OP3: delta hedging and closing the trade

An option trade also creates exposure to RTM's price direction

Buy 10 calls × 100 shares per contract × delta 0.50 = +500 delta

Sell 500 RTM shares: −500 delta. Combined portfolio delta ≈ 0.

Positive portfolio delta: sell RTM. Negative portfolio delta: buy RTM.

While the position is open

Check actual fills and portfolio delta.
Rebalance as delta changes.
Leave room inside the case limits.

When the pricing gap narrows

Close or reduce the option position.
Adjust its RTM hedge as well.
Reassess before the next weekly shock.

Costs: bid–ask spread + $0.02 per RTM share + $2 per option contract, per trade.

Deep liquidity limits price impact in this case.

Algo ETF Arb Case

Rotman

ETF arbitrage: comparing tradable prices

One RITC unit corresponds to one BULL share + one BEAR share

Basket purchase: CAD 35 ÷ 1.40 CAD per USD = USD 25.00

ETF sale: USD 25.10 **Gross difference: USD 0.10 per unit**

Illustrative executable prices, before other costs. Convert all legs to the same currency.

ETF is expensive

Sell RITC at its bid.
Buy BULL and BEAR at their asks.

ETF is cheap

Buy RITC at its ask.
Sell BULL and BEAR at their bids.

Trade only when the gap covers execution costs. Track the FX cash balance too.

Algo ETF Arb Case

Rotman

Tenders: the exit cost determines the opportunity

Institution buys 98,000 RITC from you at USD 25.15

You SELL. Starting flat, your position becomes −98,000 RITC.

Best ask: $25.08. Are all 98,000 shares available at that price?

$0.07 × 98,000 = $6,860 only if every share can be bought back at $25.08, before fees.

Before accepting

Estimate the cost across book levels.
Include fees and available capacity.
Compare buying back with conversion.

While completing the trade

Track partial fills and pending orders.
Smaller orders take longer to complete.
Limit orders may wait without filling.

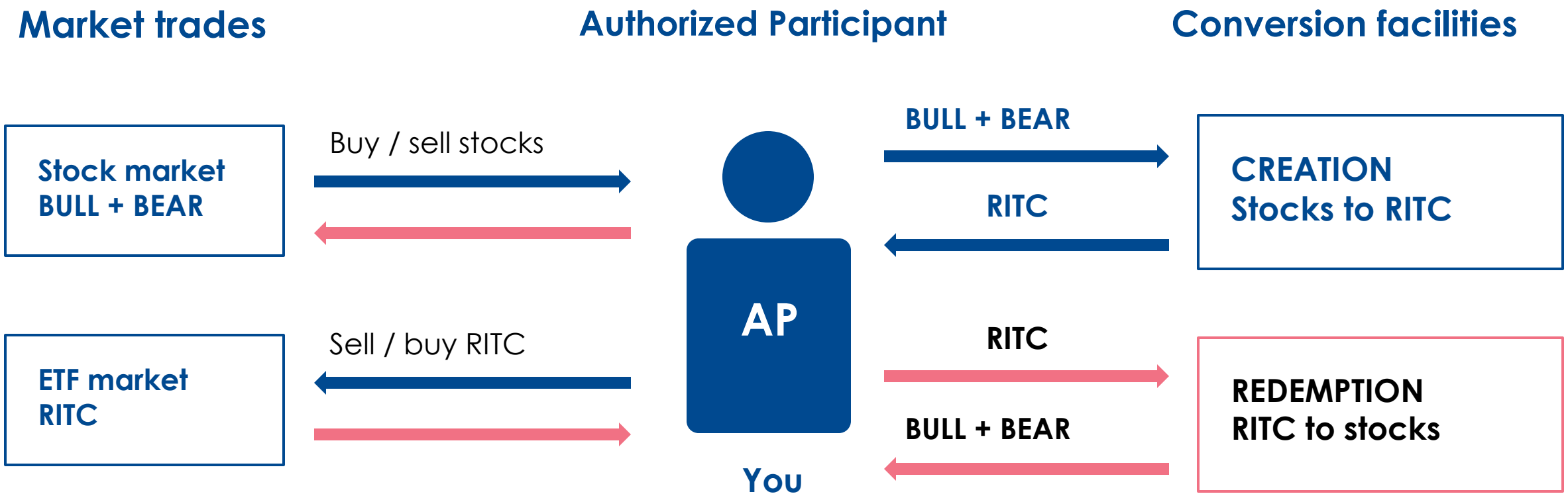
Check the tender deadline. Evaluate the full exit cost before accepting.

Plan the exit if the spread widens. A rebate or an apparent price gap does not guarantee profit.

Algo ETF Arb Case

Rotman

ETF creation and redemption in the RIT case



Blue: buy basket, create, sell RITC. Pink: buy RITC, redeem, sell basket.

10,000 BULL + 10,000 BEAR ⇄ 10,000 RITC

Each direction: US$1,500 fee

Converters: manual or API use. Market trades must be algorithmic. Arrows show securities; cash moves the other way.

Q&A

Rotman

financelab@rotman.utoronto.ca